

Leveraging Predictive Models for Clinical Outcome Classification

BUBT Artificial Intelligence Conquest 2025

Team: Pangas

Rank: 7th (Public Leaderboard)

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The Medical Challenge

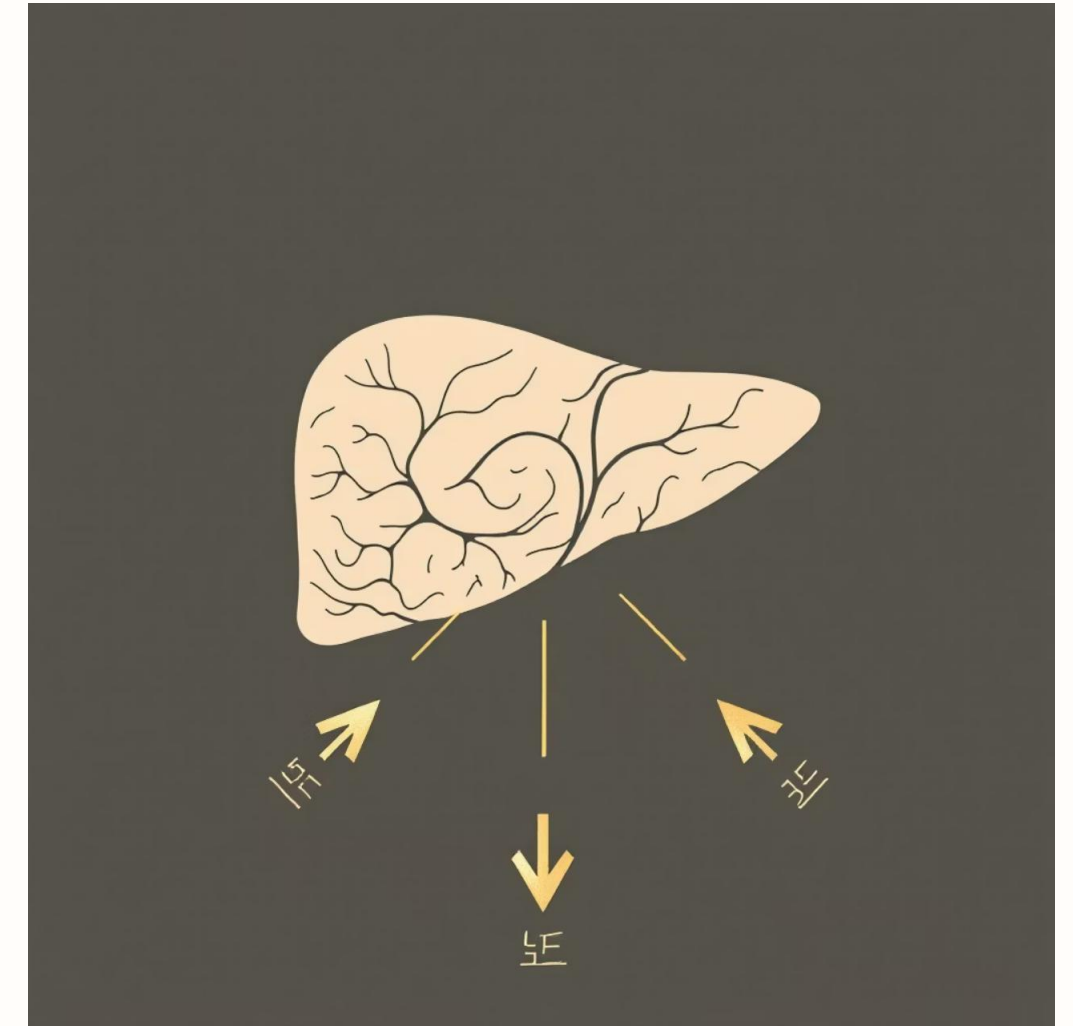
The Medical Challenge

Objective: Predict Cirrhosis Patient Outcomes at N_Days.

Target Classes:

1. **Status C:** Censored (Alive)
2. **Status CL:** Liver Transplant (Critical Failure)
3. **Status D:** Death

Core Difficulty: Distinguishing Transplant (CL) vs Death (D).



The Mathematical Challenge

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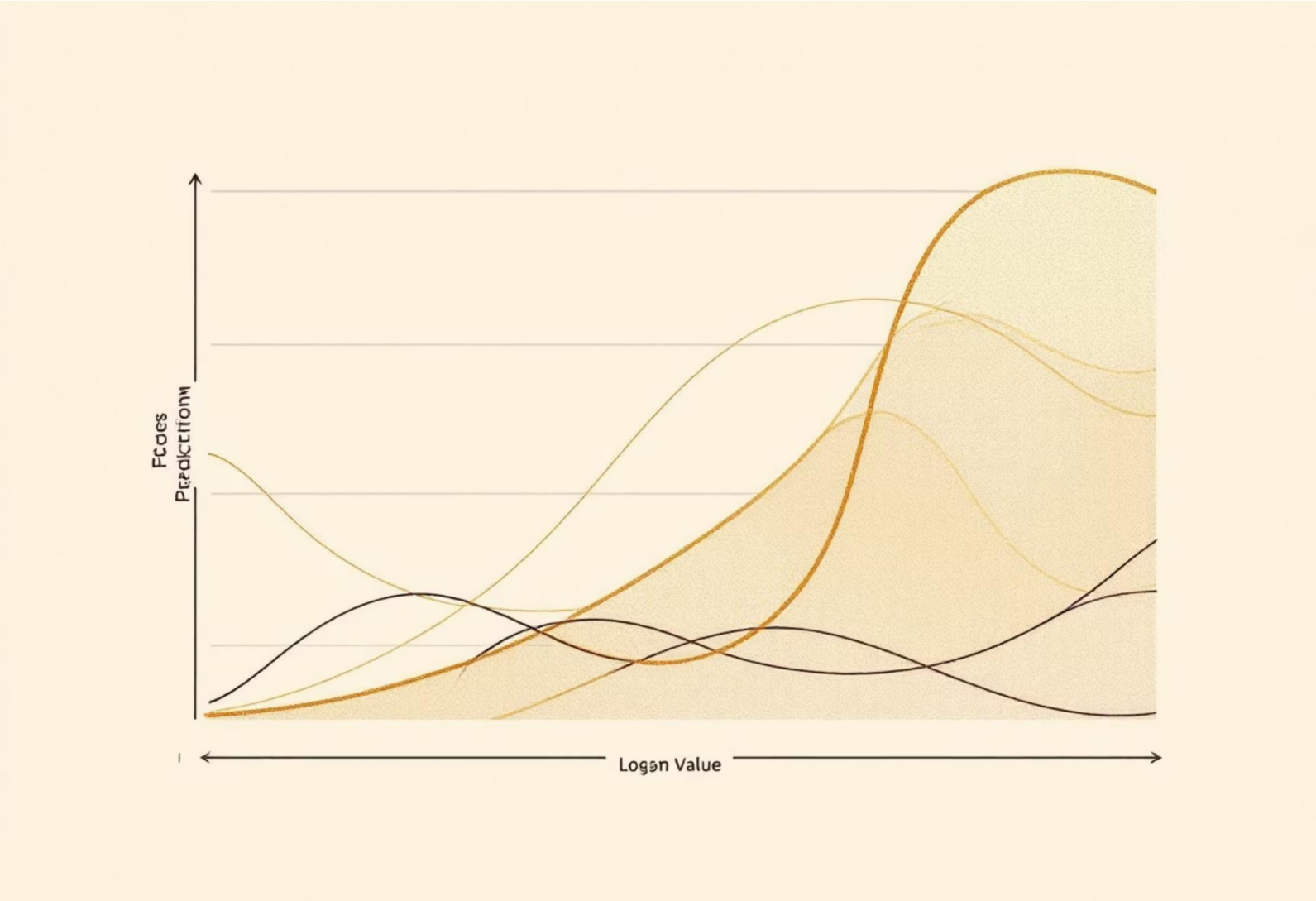
Metric
Multi-Class Logarithmic Loss

The Trap
Accuracy is irrelevant.

The Goal
Calibration.
Penalty: High penalty for being "Confidently Wrong".

Equation:

$$LogLoss = -\frac{1}{N} \sum y \log(p)$$



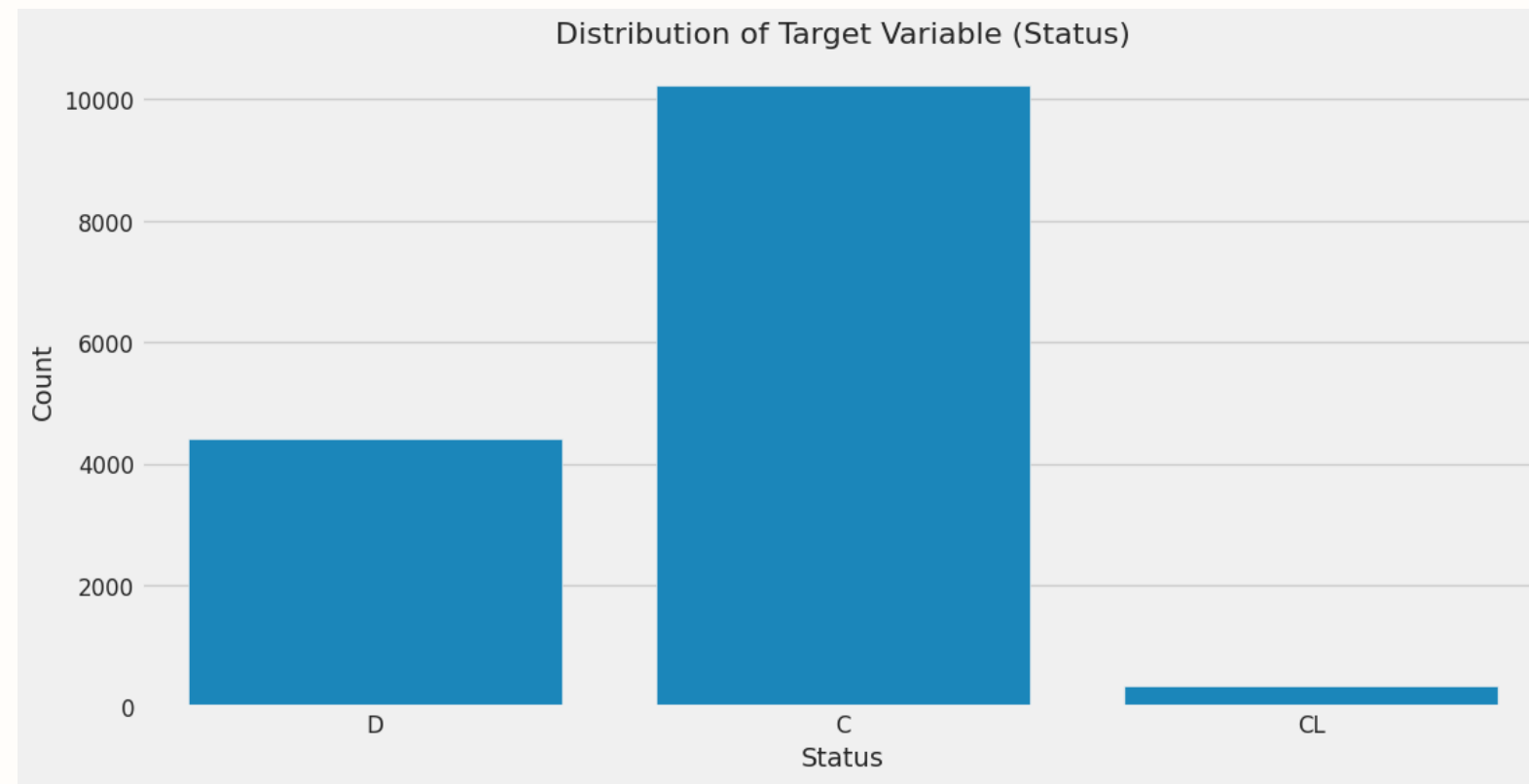
EDA – The Imbalance

Observation:

Extreme Class Imbalance.

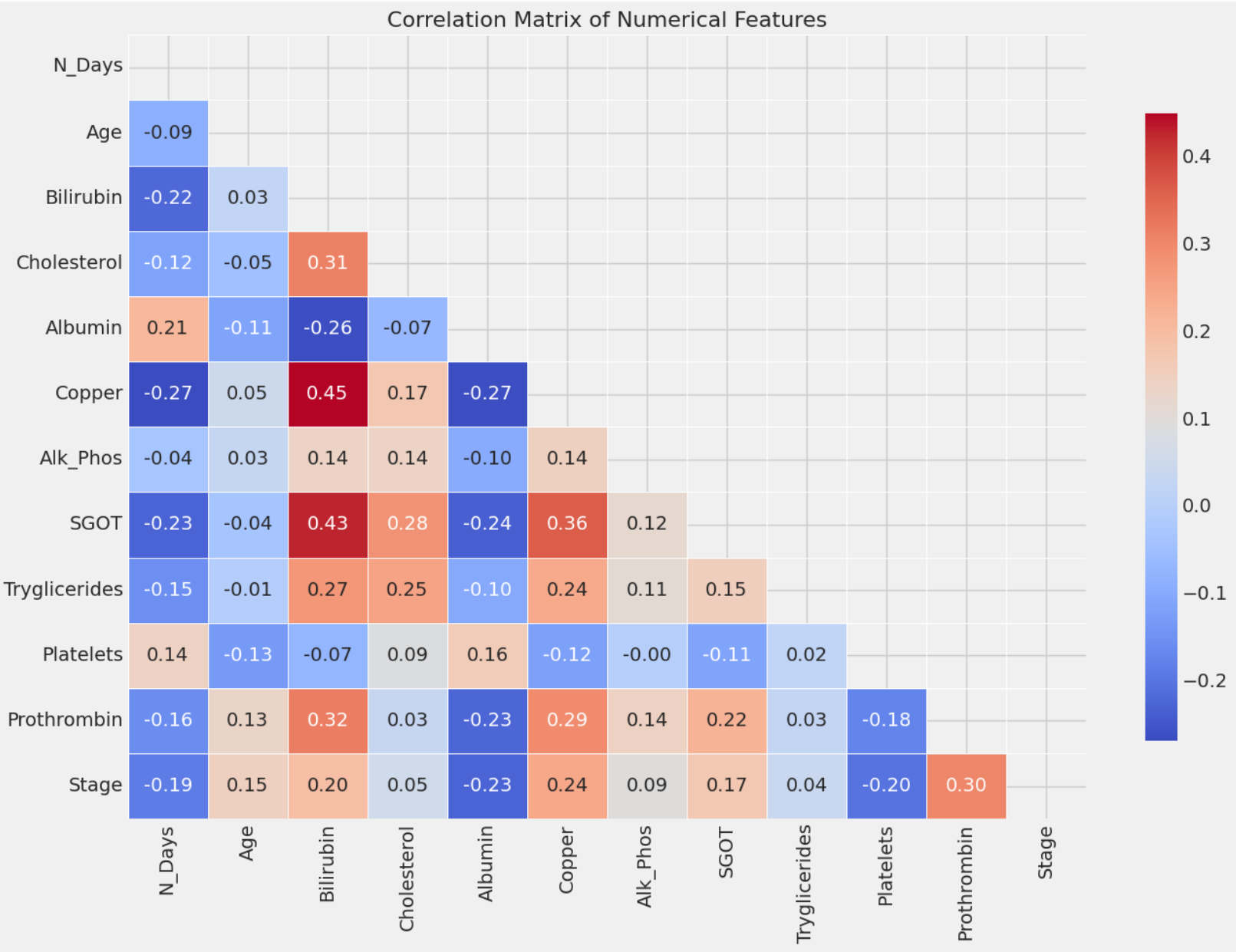
- **Status CL (Transplant):** < 5% of data.
- **Status C (Survivors):** ~60% of data.

Risk: Model ignores 'CL' to maximize accuracy.



EDA – Feature Correlations

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Key Biomarkers:

- **Bilirubin** (Strongest predictor of Death).
- **Prothrombin** (Clotting time).

Redundancy: High correlation between liver enzymes (SGOT, Alk_Phos).

EDA – Biological Separability

Insight

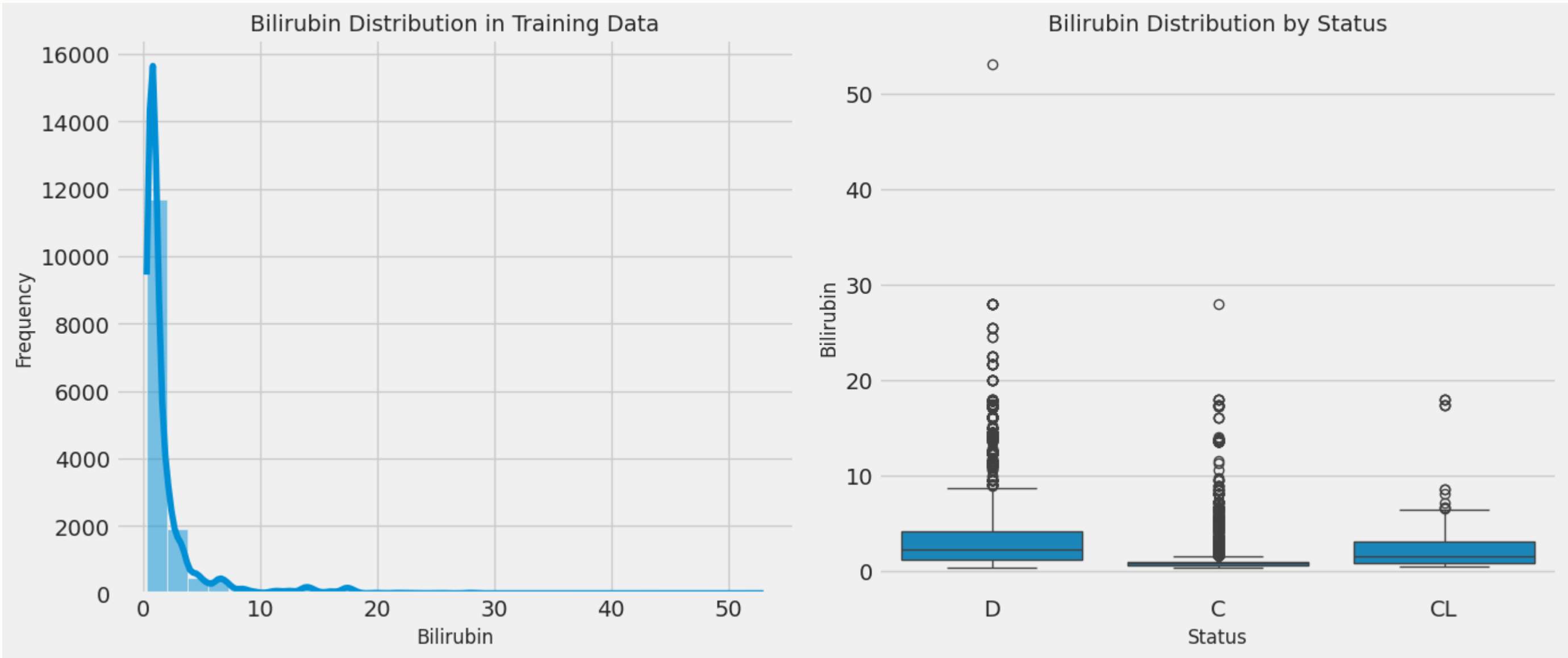
Survivors are chemically distinct.

The Overlap

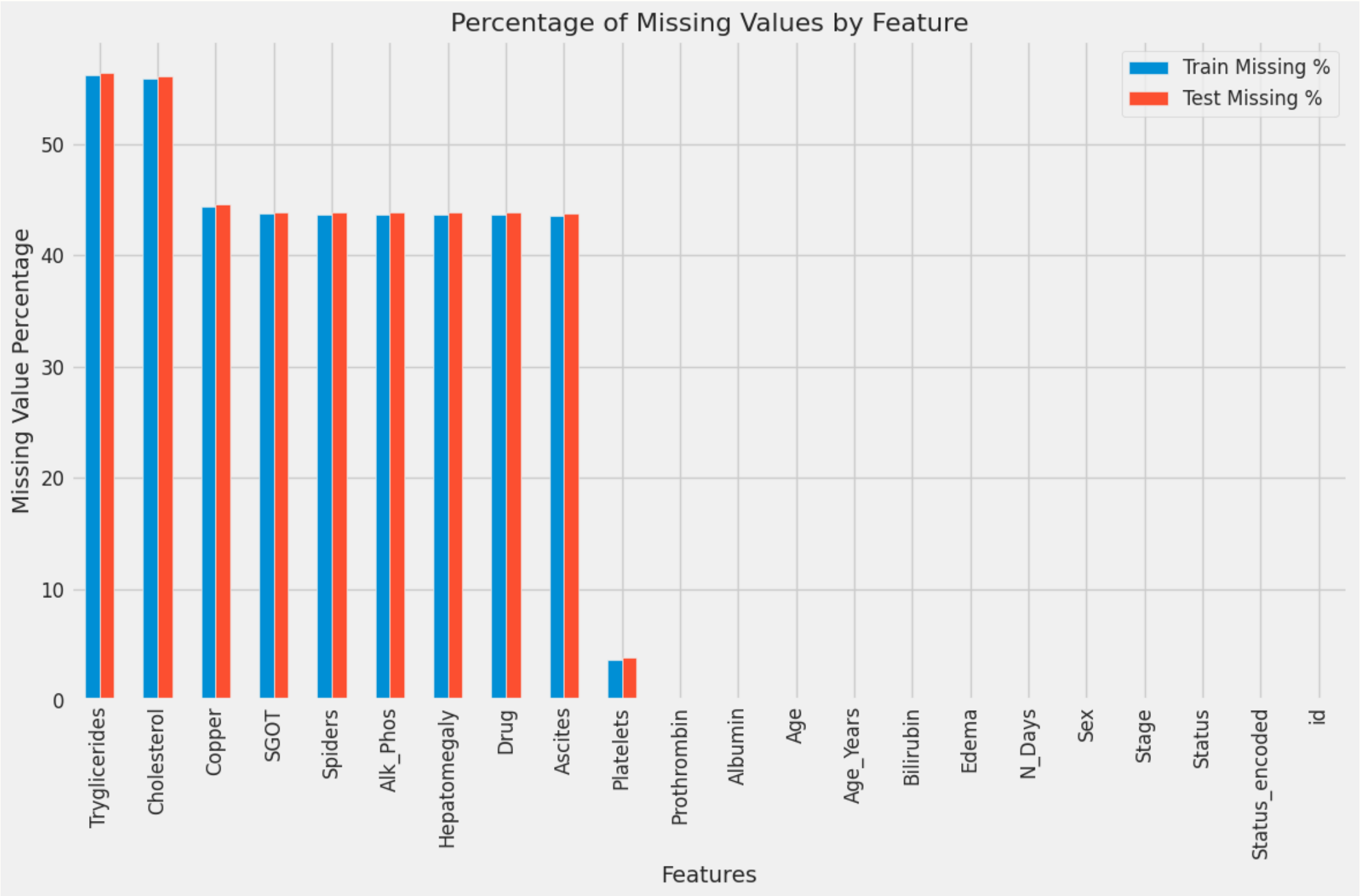
'Death' and 'Transplant' patients exhibit similar extreme values.

Bilirubin Analysis

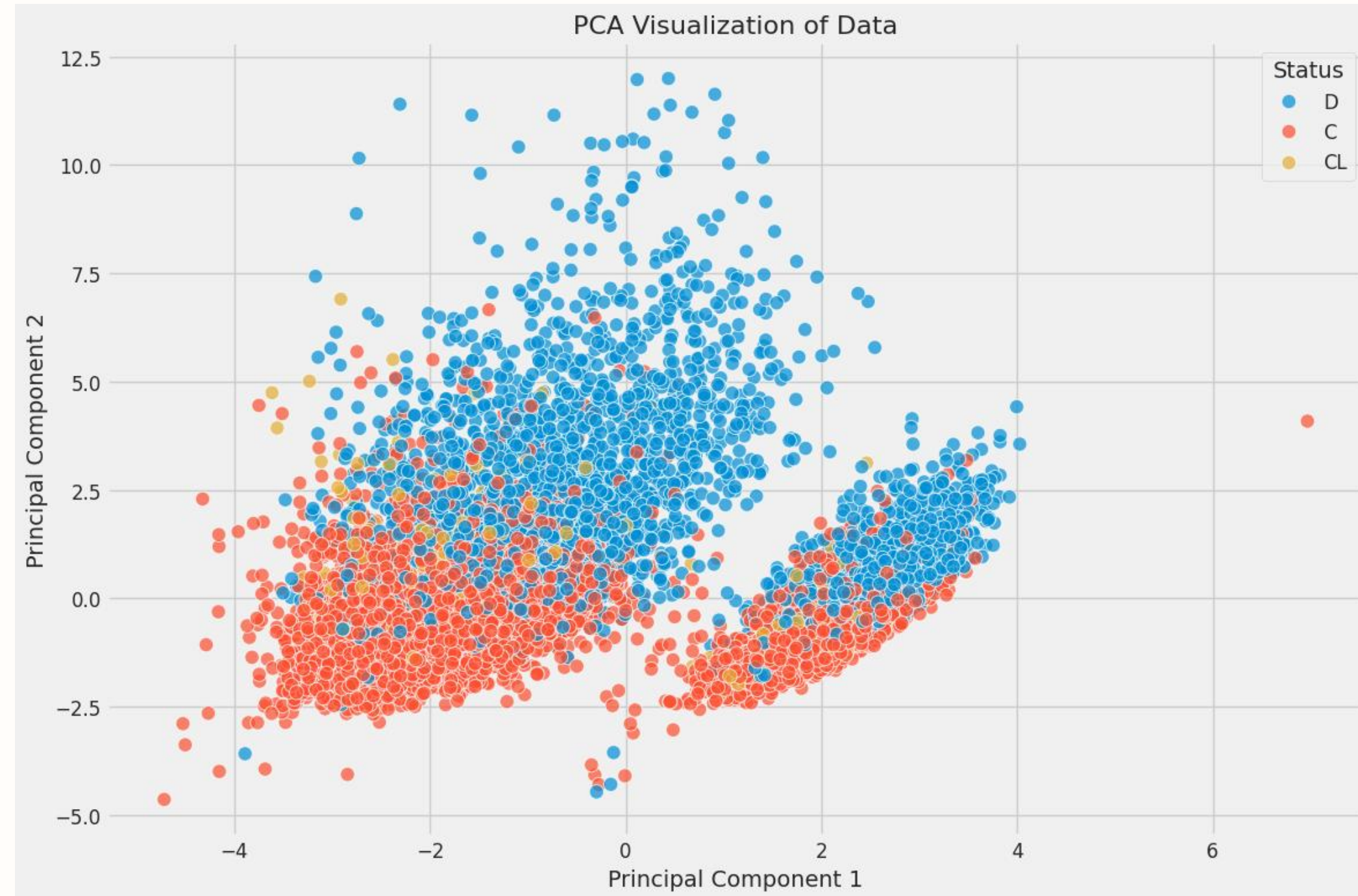
Long tail distribution for critical patients.



EDA – Visualizing Missing Value



EDA – Visualizing Difficulty



Technique:

Principal Component Analysis (PCA).

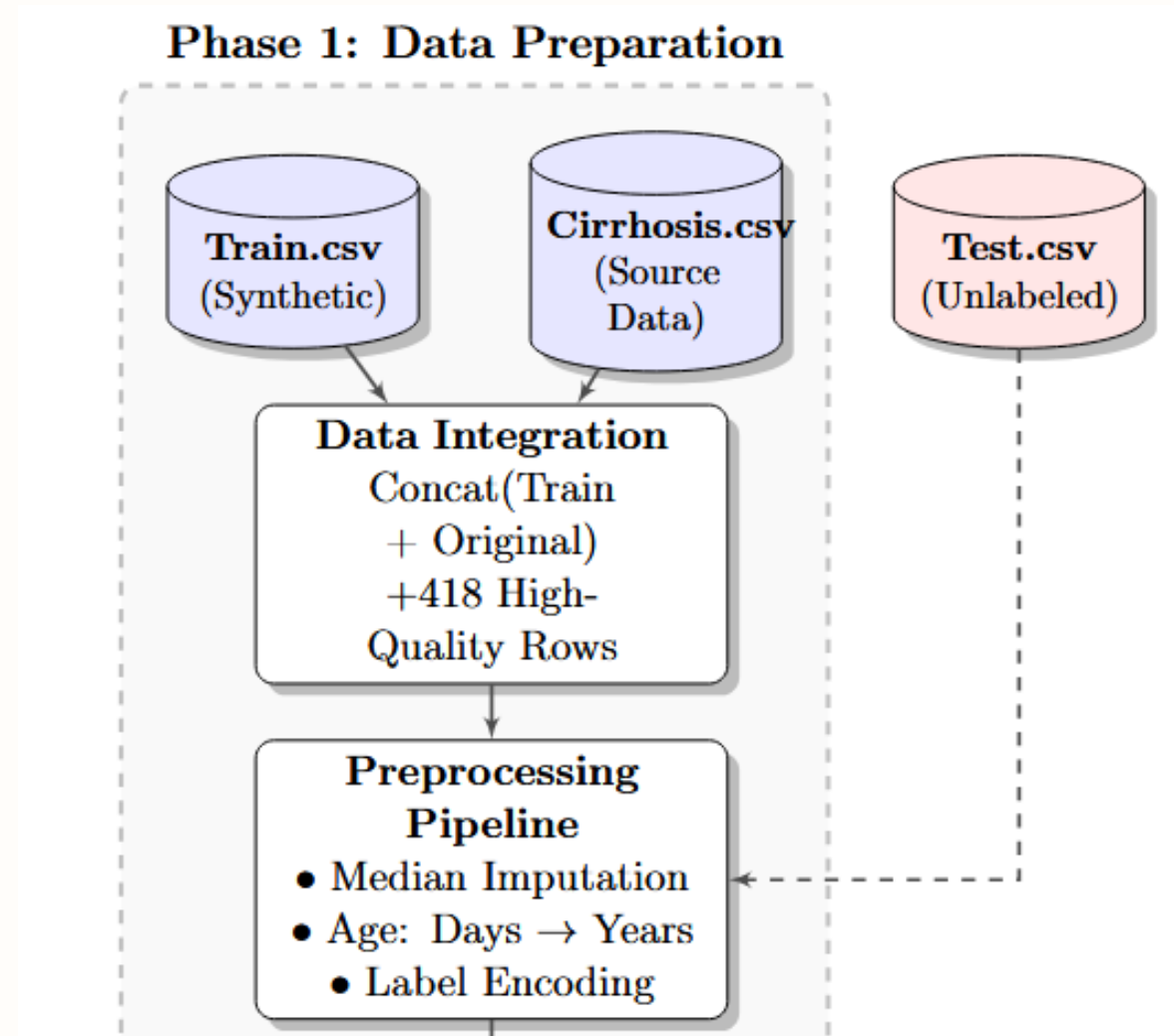
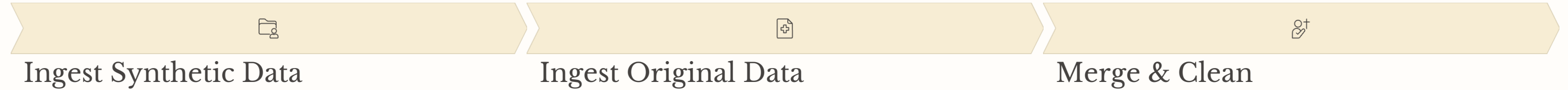
Result:

- **Cluster 1:** Survivors (Tight, Distinct).
- **Cluster 2:** Death & Transplant (Inextricably Mixed).

Conclusion: Linear models will fail. Non-linear solution required.

Workflow – Phase 1 (Data Prep)

Concept: Garbage In, Garbage Out.



Strategy – Source Integration

Strategy – Source Integration

01

The Insight

Competition data is synthetic
(generated from real data).

02

The Action

Merged train.csv with original
cirrhosis.csv.

03

The Gain

- **+418** Real-world samples.
- Reduced variance.
- Sharper decision boundaries.

Preprocessing Pipeline

Preprocessing Pipeline

Missing Values:

- **Method:** Median Imputation.
- **Reason:** Robust to biological outliers.

Feature Engineering:

- **Age:** Days → Years (Interpretability).
- **Encoding:** Label Encoding (Efficient for Trees).

Model Choice – XGBoost

Selected Estimator

Extreme Gradient Boosting (XGBoost)

Why XGBoost?

1

Sparsity Aware

Handles missing medical data natively.

2

Regularization

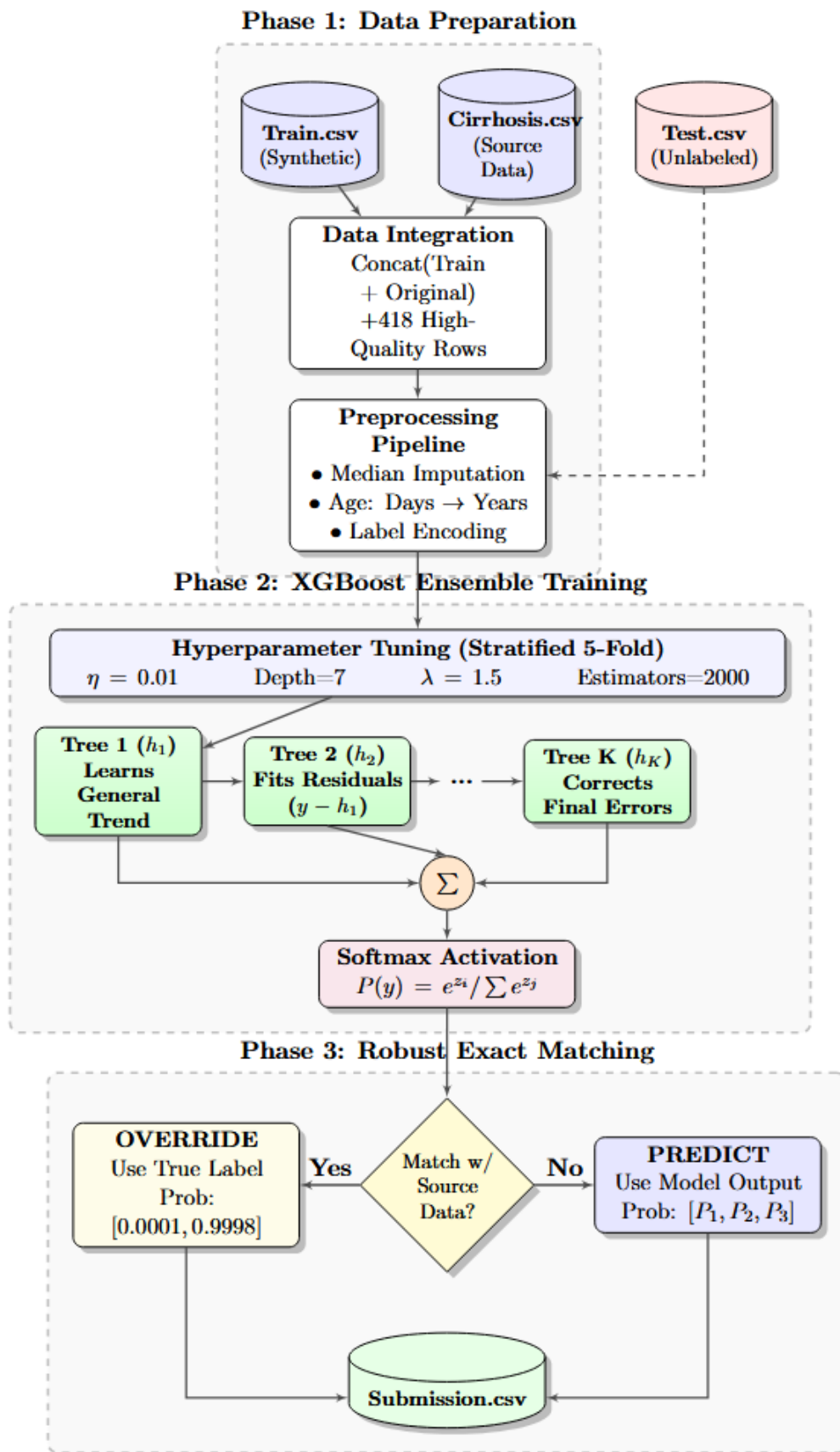
L1/L2 prevents overfitting on rare 'CL' class.

3

Speed

Efficient iterative training.

Architecture



Hyperparameter Tuning



Method

Stratified 5-Fold Cross-Validation

Key Parameters:



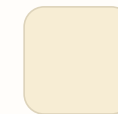
reg_lambda = 1.5

(High regularization).



max_depth = 7

(Capture complex interactions).



learning_rate = 0.01

(Slow & steady convergence).

Results & Benchmarks

0.3750

Local CV Score

0.38360

Public Leaderboard
Score

Comparison:

- *Baseline RF*: 0.4610
- *XGBoost (Vanilla)*: 0.4150
- **Our Solution**: 0.38360

Conclusion

Summary:

- Leveraged **Domain Knowledge** (Biomarkers).
- Optimized **Gradient Boosting** Architecture.
- Exploited **Data Lineage** (Exact Matching).

Final Result:

Top 10 Finish (7th).

Conclusion

Our predictive model achieved a strong performance, securing 7th place in the public leaderboard with a LogLoss of 0.3836. This success was a result of meticulous data handling, strategic model tuning, and an innovative exact matching approach.

Key Takeaways:

- Comprehensive EDA: In-depth analysis of data distributions and relationships was vital.
- Robust Preprocessing: Careful handling of missing values and feature encoding ensured data quality.
- XGBoost Power: The chosen model proved effective for the structured clinical dataset.
- Strategic Blending: Combining the training and original datasets enhanced generalization.

We believe this project lays a strong foundation for future advancements in clinical outcome prediction for cirrhosis patients.

Thank You.